Network

Reconnaissance

From Stress to Distress: A Personalized Detection Approach Using Gaussian Process Regression

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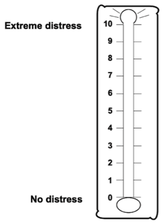
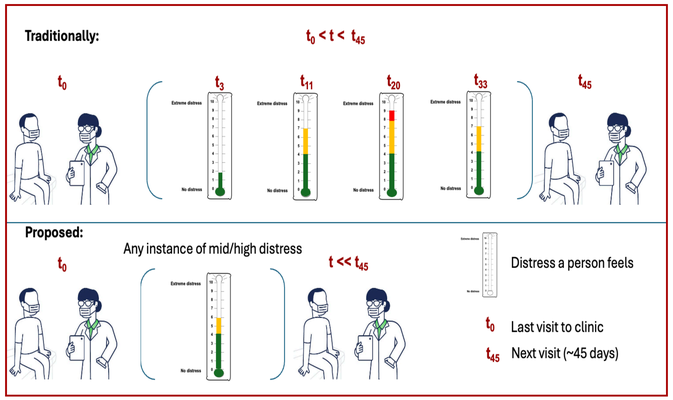
INTRODUCTION



- **Distress** is a multifactorial experience that impairs one's ability to cope with cancer and its treatment
  - Example of symptoms: **stress**, fear, and anxiety
- The National Comprehensive Cancer Network (NCCN) Distress Thermomether is effective in assessing distress from cancer patients[2]



- Distress screenings traditionally occur in clinical visits
  - The occurence of these visits vary, allowing distress to persist between assessments[1]
- To detect distress, we propose using **Gaussian Process Regression (GPR)**
  - Its non-parametric property and uncertainty measure provides a promising approach towards a personalized model for distress dection
- To validate feasibility, we evaluate GPR on the **WESAD dataset**[3], where stress serves as a proxy for distress

METHODS

Wearable Stress and Affect Detection (WESAD) Dataset

The WESAD dataset contains wrist- and chest-worn sensor data from 15 participants across neutral, stress, and amusement phases. Each participant completed a self-report questionnaire at the end of each phase

Subjective Data

PANAS questionnaire, STAI questionnaire, SAM questionnaire, SSSQ questionnaire

- The Positive and Negative Affect Schedule (PANAS) questionnaire was used to find the correlation between distress and stress responses
  - Responses ranged from 1 (Not at all) to 5 (Extremely)
  - Evaluated correlation with Pearson Correlation Coefficient

Objective Data

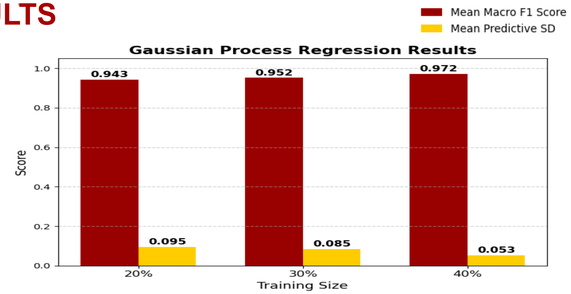
Blood volume pulse, **electrocardiogram**, electrodermal activity, electromyogram, respiration, body temperature, and three-axis acceleration

- Electrocardiogram (ECG) signals from the chest sensor were used to derive heart rate and heart rate variability as input features
- For each participant, a **subject-specific GPR model** was trained to predict stress
  - Neutral and stress data were labeled 0 and 1, respectively, with a classification cutoff of 0.5

RESULTS

Emotion	Correlation to Distress	Correlation to Stressed
Ashamed	0.829	0.846
Annoyed	0.825	0.792
Frustrated	0.823	0.797
Scared	0.804	0.893
Nervous	0.773	0.878
Jittery	0.727	0.719
Active	0.644	0.788
Afraid	0.628	0.859
Happy	-0.602	-0.667

- A strong positive correlation (r = **0.786**) between stress and distress
- Additionally, several items from the PANAS questionnaire were correlated with both stress and distress



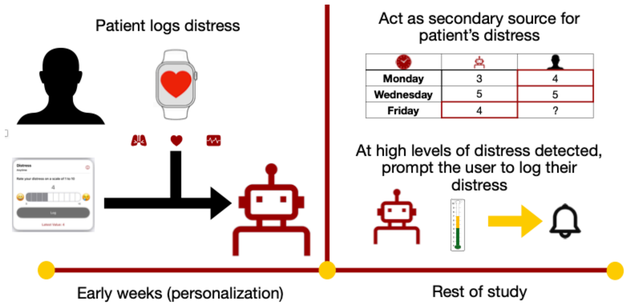
- Results indicate that GPR can quickly personalize while maintaining prediction reliability
- Amusement readings were used to evaluate uncertainty, resulting in a mean predictive SD of 0.405. This high value indicates greater uncertainty from the model

CONCLUSION

- This study demonstrates the feasibility of GPR for personalized stress detection
  - Establishes GPR as a foundational method for distress detection through stress–distress correlation
- GPR offers reliable performance
  - Enables fast deployment from limited training data
  - Captures predictive uncertainty to avoid false positives

FUTURE WORK

- An IRB-approved study is in progress in which cancer patients will report their distress score and objective health data is collected via smartwatches
  - Implement in Assuage[4], a HIPPA-compliant, offline-first IOS application
- This will enable the development of real-time, personalized distress detection to enhance cancer patient care



Install Demo via **TestFlight (iOS only)**

REFERENCES

[1] Kristin K Ownby. "Use of the distress thermometer inclinical practice". In: Journal of the advanced practitioner in oncology 10.2 (2019), p. 175.

[2] Michelle B Riba et al. "Distress management, version 3.2019, NCCN clinical practice guidelines in oncology".In: Journal of the National Comprehensive Cancer Network 17.10 (2019), pp. 1229–1249

[3] Philip Schmidt et al. "Introducing wesad, a multimodal dataset for wearable stress and affect detection". In: Proceedings of the 20th ACM international conference on multimodal interaction. 2018, pp. 400–408.

[4] Alyssa Donawa et al. "Designing Survey-Based Mobile Interfaces for Rural Patients With Cancer Using Apple's ResearchKit and CareKit: Usability Study". In: JMIR Formative Research 8.1 (2024), e57801.

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